

various lightweight IoT protocols for connecting devices, such as CoAP and MQTT. Kaa can support any open IoT protocol. This platform allows the building of all those applications that work over any network connection, be it persistent or intermittent. We may select one of the existing transport protocols that come with Kaa, or can even create a customised transport protocol and plug it into the system. Kaa uses MQTT as the default protocol, and is built on top of CoAP and MQTT. It defines all the basic rules of communication between devices and the platform. This protocol is open and allows for any arbitrary message formats.

2. **Management of devices:** Kaa facilitates the registration of digital twins, which represent devices, and other such entities managed by the platform. It also allows the storage of different device attributes, which provide detailed information about any feature of the device. Such attributes can be the MAC address, serial number, location, etc. Apart from simple data types, attributes can also contain more complex and structured objects, like a list of connected peripherals with their respective properties. Besides, we can construct different filters using the device attributes to segment the device population into sets of individually managed groups. Kaa automatically re-evaluates filters whenever the device attributes change. To connect to the Kaa platform, a device just has to present valid credentials, such as tokens, pre-shared keys, login and password combinations, etc. Kaa tracks any connected device throughout its life cycle.
3. **Data collection:** Kaa provides an easy-to-use protocol to collect data from different connected devices. This protocol also ensures reliable data delivery along with their respective response codes indicating the outcome of data processing by this platform. Once the data is received by the platform, it can be sent to multiple processing pipelines. In case of any error in the midst of processing, like processor overload or disk crash, the device is notified about it. Hence the device always knows whether the data submitted is safe to delete or is required to be resent. This protocol supports batching to minimise network usage and increase the data throughput. It enables devices to buffer data locally before it is uploaded in a single message. Besides improving the network efficiency, this feature is useful in IoT deployments when connectivity is intermittent and hence helps in preserving the device's battery life. It allows the collection of both types of data sets—structured and unstructured. It can be categorised into primitive types, like plain numbers, text or compounds like key-value arrays, maps or nested objects.
4. **Command execution:** The command execution feature of Kaa allows the delivery of messages with arbitrary payload to any of the connected devices, and also permits the receipt of near real-time responses from them. For instance, one can remotely check the present temperature using a home thermostat, simultaneously point a security camera to any defined area, open any vehicle's trunk, and more. Kaa uses two-way communication that allows different devices connected to it to send responses back to the server. For lightweight commands, one can use the synchronous option, which means that the caller will be put on hold until the results of command execution are delivered. For resource-consuming commands, we may choose the asynchronous option, which enables our platform to notify its caller about the command execution result.
5. **Analysis and data processing:** Kaa gives a lot of freedom when processing the collected data. This platform supports data collection adapters which allow sending data to different databases and data analytics systems. As Kaa has got a highly modular architecture, any new integration to it requires minimum effort to update or change an existing adapter, or to create a custom one. Raw or unstructured data can also be converted to a well-structured time series that is convenient for pattern analysis, analytics, charting, visualisation, etc. Similarly, the time series can also be pushed into an analytics or processing system of choice for any raw data set. Working with time series is really very flexible in Kaa. It not only displays the main value for a time series, like temperature, it also allows its users to set up tags, which allow viewing some additional data as well, like light intensity, humidity, etc. Tag values are actually fetched from the set of collected raw data and then attached to each data point in a time series. Also, it is possible to develop multiple time series from one data sample.
6. **Data visualisation:** The data visualisation part of Kaa comprises a rich set of widgets, such as charts, gauges, tables, maps, etc. We can use all these widgets to analyse and visualise different types of data, whether statistics, telemetry, geolocation, metadata, or any other type—historical as well as current. All of these widgets are configurable and allow modification of their data sources and visual representation. To address any special use cases, the visualisation component of Kaa allows easy plugging into custom widgets. Apart from data visualisation, widgets allow interaction with various devices by changing configurations and metadata, sending commands, etc. Dashboards allow widgets to be organised into sets of logical groups and to define their layout. Kaa supports dashboard templating, allowing the reuse of one configuration for more than one device's dashboard. Kaa can also be integrated to any data visualisation tool of choice.
7. **Configuration management:** Configuration management is very useful for controlling the behaviour of devices and managing various data processing parameters—feature flagging, edge analytics and other such